

Open Source Backup Solutions that You Must Have

Backup solutions are a must for enterprises. Without a backup and recovery solution, businesses could lose a fortune if there is a data breach or loss due to some accident. This article offers a few backup and restore solutions which, true to the tradition of open source, are free of cost.



According to a recent Forbes report, 2.5 quintillion bytes of data are being generated every day and this is expected to multiply several times in the coming years as we inch towards the AI-enabled smart digital era. All the new technologies like machine learning and IoT are fuelling data creation and, at the same time, increasing our vulnerability to data breaches. Well, the spectre of potential breaches is definitely haunting both small and large businesses. Losing data can affect businesses very hard. For most medium to large scale companies, experiencing even a single hour of downtime can cost more than US\$ 10,000 while for smaller firms, the cost can go up to US\$ 1 million. With such potential losses, it's little wonder that different enterprises invest heavily in backup and recovery solutions to help them get their data back.

If we look at a small business or home user, a hard drive crash may lead to the loss of decades of family photos or important business reports, tax forms and other such irreplaceable files. Without any backup and recovery solution, it can take days or even years to repair such damages. A backup solution helps businesses to prevent the heavy losses that can be caused by various uncontrollable disasters.

Backup and recovery is generally the process of creating and storing copies of the data, which can be used to protect firms and organisations against data loss. This is also referred to as operational recovery. Recovery using a backup typically involves restoring the data to its original location or to any alternate location, where it can be used in place of the lost or damaged

data. A proper backup copy is stored in a different system or medium (like tape) from that of the primary data to protect against any possibility of data loss in case of primary hardware or software failure. Storing the copy of data on a separate medium is critical to protect against any primary data loss or corruption. This additional data storage medium can be as simple as a USB stick, an external drive, or something more substantial, like a cloud storage container, disk storage system, or tape drive. We can have the alternate medium in the same location as that of the primary data or even at a remote location. It is recommended that the backup copies are made on a regular basis in order to minimise the data lost between consecutive backups. The longer the time interval between consecutive backup copies, the higher the possibility for data loss while recovering from a backup.

Fortunately, the open source community has many solutions available that can support individuals and organisations of all sizes to protect their valuable data, with minimal expense.

Why do we need open source backup solutions?

1. The main purpose of the backup is to create and store a copy of the data that can be easily recovered in case of any primary data failure, which can occur due to data corruption, hardware or software failure, or even a human-caused event like malicious malware attacks, or accidental deletion of data.
2. Backup copies allow data to be easily restored from an earlier point in time to help the business recover from an unexpected event.